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POSITION DETECTION DEVICE

Abstract

A position detection device includes a substrate, a transmitter coil formed in the substrate, and first and second receiver coils formed in the substrate and inductively coupled by electromagnetic induction caused by energizing the transmitter coil. The substrate is a multilayer substrate with six or more wiring layers and insulating layers alternately arranged between each of the wiring layers. The substrate includes a plurality of conductive penetrating portions each formed to penetrate at least one of the insulating layers so as to connect the six or more wiring layers. Each of the first and second receiver coils is wound a plurality of turns in each wiring layer, excluding at least one wiring layer, among the six or more wiring layers, and is connected through the conductive penetrating portions. At least one of the conductive penetrating portions is disposed inside each of the first and second receiver coils.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application of International Patent Application No. PCT/JP2023/038570 filed on Oct. 25, 2023, which designated the U.S. and claims the benefit of priority from Japanese Patent Application No. 2022-187596 filed on Nov. 24, 2022. The entire disclosures of all of the above applications are incorporated herein by reference.

TECHNICAL FIELD
[0002] The present disclosure relates to a position detection device.
BACKGROUND
[0003] Conventionally, there has been known an angular position sensor that detects a position of a detection object that is rotatable.
SUMMARY
[0004] A position detection device according to an aspect of the present disclosure includes a substrate, a transmitter coil formed in the substrate, and first and second receiver coils formed in the substrate and inductively coupled by electromagnetic induction caused by energizing the transmitter coil. The substrate is a multilayer substrate with six or more wiring layers and insulating layers alternately arranged between each of the wiring layers. The substrate includes a plurality of conductive penetrating portions each formed to penetrate at least one of the insulating layers so as to connect the six or more wiring layers. Each of the first and second receiver coils is wound a plurality of turns in each wiring layer, excluding at least one wiring layer, among the six or more wiring layers, and is connected through the conductive penetrating portions. At least one of the conductive penetrating portions is disposed inside each of the first and second receiver coils.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0005] Objects, features and advantages of the present disclosure will become apparent from the following detailed description made with reference to the accompanying drawings. In the drawings:
[0006] FIG. **1** is a block diagram of an electrification system configured using a position detection device of a first embodiment;
[0007] FIG. **2** is a diagram showing a relationship between the position detection device and a drive unit;
[0008] FIG. **3** is a top view of a rotating flat plate and the position detection device;
[0009] FIG. **4** is a perspective view of the position detection device;
[0010] FIG. **5** is a cross-sectional view of the position detection device taken along line V-V of FIG. **4**;
[0011] FIG. **6** is a plan view of a printed circuit board according to the first embodiment, as seen in the direction indicated by arrow VI in FIG. **5**;
[0012] FIG. **7** is a block diagram of the position detection device;
[0013] FIG. **8** is a cross-sectional view of the printed circuit board according to the first embodiment;

[0014] FIG. **9** is a diagram for explaining the configuration of wiring layers, a first receiver coil, and vias in the printed circuit board of the first embodiment;
[0015] FIG. **10** is a diagram for explaining a connection of the first receiver coil formed in each wiring layer in the printed circuit board of the first embodiment;
[0016] FIG. **11** is a diagram for explaining the configuration of wiring layer, a first receiver coil, and vias in a printed circuit board of a comparative example;
[0017] FIG. **12** is a diagram for explaining an area in which vias can be formed in the printed circuit board of the comparative example;
[0018] FIG. **13** is a diagram for explaining the configuration of wiring layers, a first receiver coil, and vias in a printed circuit board of a second embodiment;
[0019] FIG. **14** is a view corresponding to FIG. **6** of the printed circuit board of the second embodiment;
[0020] FIG. **15** is a diagram for explaining the configuration of wiring layers, a first receiver coil, and vias in a printed circuit board of a third embodiment;
[0021] FIG. **16** is a view corresponding to FIG. **6** of the printed circuit board of the third embodiment; and
[0022] FIG. **17** is a view corresponding to FIG. **6** of a printed circuit board of a fourth embodiment.

DETAILED DESCRIPTION

[0023] Next, a related art is described only for understanding the following embodiments. An angular position sensor according to the related art has two excitation coils and two sensing coils. The angular position sensor generates a magnetic field between the two excitation coils and the detection object by supplying current to the two excitation coils, and detects a rotation angle of the detection object based on a detection signal generated by the change in the magnetic field. The two excitation coils and the two sensing coils are formed on a four-layered substrate. Specifically, of the two excitation coils, one excitation coil is formed on first and second layers, and the other excitation coil is formed on third and fourth layers.

[0024] Moreover, each of the two sensing coils is formed in series from the first layer through the second and third layers to the fourth layer. Each of the two sensing coils has a portion formed by winding four turns in a clockwise or counterclockwise direction in each layer, for a total of sixteen turns. Each of the two sensing coils has a portion that is wound clockwise or counterclockwise in each of the four layers, and the portions are electrically connected through one of the three vias that penetrate from the first layer to the fourth layer of the multilayer substrate.

[0025] The inventors have investigated a position detection device that uses a transmitter coil serving as an excitation coil and a receiver coil serving as a sensing coil, and that allows the total number of turns of the receiver coil to be increased. The total number of turns of the receiver coil is the sum of the total number of turns of the receiver coil wound on each layer of a multilayer substrate. As a method to increase the total number of turns of the receiver coil, there is a method of increasing the number of layers of the multilayer substrate, as well as increasing the number of turns of the receiver coil formed in each layer of the multilayer substrate with increased layers.

[0026] However, according to the diligent study by the inventors, forming through-conductive parts such as vias that penetrate all layers of the multilayer substrate leads to a reduction in area available for forming the receiver coil in each layer due to the through-conductive parts. This limits the number of turns of the receiver coil in each layer, and it may not be possible to increase the total number of turns of the receiver coil to the desired number.

[0027] According to one aspect of the present disclosure, a position detection device includes a substrate, a transmitter coil formed in the substrate, and a first receiver coil and a second receiver coil formed in the substrate and inductively coupled by electromagnetic induction caused by energization to the transmitter coil. The substrate is a multilayer substrate with six or more wiring layers and insulating layers alternately arranged between each of the six or more wiring layers. The substrate includes a plurality of conductive penetrating portions each of which is formed to

penetrate at least one of the insulating layers so that the plurality of conductive penetrating portions connect the six or more wiring layers. Each of the first receiver coil and the second receiver coil is wound a plurality of turns in each wiring layer, excluding at least one wiring layer, among the six or more wiring layers, and is connected through the plurality of conductive penetrating portions. At least one of the plurality of conductive penetrating portions is disposed inside each of the first receiver coil and the second receiver coil.
[0028] According to this configuration, compared to a case where the first receiver coil and the second receiver coil are formed on all six or more wiring layers, the number of conductive penetrating portions for connecting the first receiver coil and the second receiver coil can be reduced. Therefore, in each of the wiring layers in which the first receiver coil and the second receiver coil are formed, the area available for forming the first receiver coil and the second receiver coil can be increased.
[0029] Hereinafter, embodiments of the present disclosure will be described with reference to the drawings. In the following embodiments, parts that are the same as or equivalent to parts described in the preceding embodiment are denoted by the same reference numerals, and description thereof may be omitted. When only a part of components is described in the embodiment, components described in the preceding embodiment can be applied to other parts of the components. The following embodiments may be partially combined with each other even if such a combination is not explicitly described as long as there is no disadvantage with respect to such a combination.
First Embodiment
[0030] The present embodiment will be described with reference to FIGS. **1** to **12**. In the present embodiment, as a position detection device, a position detection device that detects rotation of a detection object will be described as an example. In the present embodiment, an example in which the position detection device is applied to an electrification system mounted on a vehicle will be described.
[0031] [Electrification System]
[0032] The electrification system, as shown in FIG. **1** for example, includes an actuator **1**, a gear **2**, a drive unit **3**, an ECU (abbreviation of Electronic Control Unit) **4**, and a position detection device S**1**. The electrification system is operated as follows. That is, the actuator **1** is controlled by the ECU **4**, and rotates the gear **2** under the control of the ECU **4**. The drive unit **3** includes a detection object to be described later, and includes components operated by the rotation of the gear **2**. The position detection device S**1** detects a displacement of the detection object provided in the drive unit **3**, and outputs a detection signal to the ECU **4**. In the present embodiment, as will be described later, the detection object includes a rotating flat plate **30**, and a rotation angle of the rotating flat plate **30** is output to the ECU **4**. The ECU **4** controls the actuator **1** in view of the detection signal from the position detection device S**1**.
[0033] Next, a configuration of the drive unit **3** in which the position detection device S**1** is disposed will be described. In the present embodiment, an example in which the position detection device S**1** is disposed in a motor such as a main motor and an in-wheel motor will be described.
[0034] The drive unit **3** is assumed to be, for example, a rotor for a motor, and as shown in FIG. **2**, includes a shaft **10** as a rotating shaft, the rotating flat plate **30**, a fixing base **40**, and the like. These components **10**, **30**, **40** are arranged coaxially with an axial direction Da of the shaft **10** as the center. In the following description, the axial direction Da of the shaft **10** will be simply referred to as the axial direction Da. In addition, for ease of viewing, FIG. **2** shows a diagram in which a transmitter coil **110**, a first receiver coil **120**, and a second receiver coil **130**, which are described below and which constitute the position detection device S**1**, are omitted.
[0035] The shaft **10** is, for example, a drive shaft, and is made of a cylindrical member. The shaft **10** is disposed so that a tire is provided on one end side and the other end side opposite to the one end side becomes the vehicle body side. For example, in FIG. **2**, the upper side of the paper corresponds to the one end side of the shaft **10**, and the lower side of the paper corresponds to the

other end side of the shaft **10**. Although details are omitted, for example, a rotating wheel and a bearing member (not shown) are attached to the shaft **10**, and the rotating wheel is supported by the bearing member in a rotatable state.

[0036] The rotating flat plate **30** is made of metal, and has an annular plate shape having a through hole **30***a*. As shown in FIG. **3**, the rotating flat plate **30** of the present embodiment has a plurality of depression portions **31** provided in an outer edge thereof at an equal interval in a circumferential direction. In other words, the rotating flat plate **30** has a configuration in which a plurality of protrusion portions **32** are disposed along the circumferential direction in the outer edge portion. That is, in the rotating flat plate **30**, a depression and protrusion structure **33** having the depression portions **31** and the protrusion portions **32** is provided along the circumferential direction in the outer edge portion.

[0037] As shown in FIG. **2**, the rotating flat plate **30** is fixed to the shaft **10** with one end of the shaft **10** inserted into the through hole **30***a* so as to rotate with the rotation of the shaft **10**. In the present embodiment, the rotating flat plate **30** corresponds to the detection object.

[0038] The fixing base **40** has a plate shape having a through hole **40***a*. The other end of the shaft **10** is inserted into the through hole **40***a* of the fixing base **40**, and the rotating wheel (not shown) is disposed in a rotatable state. On the fixing base **40**, the position detection device S**1** is disposed to face the protrusion portion **32** of the rotating flat plate **30** in the axial direction Da. As shown in FIG. **3**, the position detection device S**1** is disposed so as to have a predetermined gap (that is, a predetermined distance) d between the position detection device S**1** and the rotating flat plate **30**.

[Position Detection Device]

[0039] Next, a configuration of the position detection device S**1** of the present embodiment will be described. As shown in FIGS. **4** and **5**, the position detection device S**1** of the present embodiment includes a printed circuit board **100** having one surface **100***a* and the other surface **100***b*. In the position detection device S**1**, a circuit board **200** and terminals **400** are disposed on one surface **100***a* side of the printed circuit board **100**, and these are integrally sealed by a sealing member **500**. In the following description, the normal direction Ds to a surface direction of the printed circuit board **100** will be simply referred to as the normal direction Ds. When the position detection device S**1** is disposed on the fixing base **40**, the normal direction Ds of the printed circuit board **100** coincides with the axial direction Da. Although not particularly shown, various electronic components such as capacitors and resistors are appropriately disposed in the printed circuit board **100**.

[0040] The printed circuit board **100** of the present embodiment has an arc plate shape. Specifically, the printed circuit board **100** is configured to coincide with an arc of a virtual circle formed around the shaft **10**. That is, the printed circuit board **100** has a shape in which a virtual circle having the printed circuit board **100** as an arc coincides with a circle formed around the shaft **10**.

[0041] As shown in FIG. **6**, the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130** are disposed in the printed circuit board **100**. As shown in FIG. **7**, the printed circuit board **100** also has a connection wire **150** that connects the circuit board **200** to the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130**. It should be noted that, in FIG. **5**, the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130** are shown in a simplified manner.

[0042] Specifically, the printed circuit board **100** of the present embodiment is a multilayer substrate in which insulating layers **101** and wiring layers **102** are alternately laminated, as shown in FIG. **8**. The printed circuit board **100** of the present embodiment has a six-layer through substrate in which insulating layers **101** and wiring layers **102** are alternately laminated. The printed circuit board **100** is composed of a ten-layer build-up substrate in which two insulating layers **101** and two wiring layers **102** are further laminated on both of one side and the other side of the through substrate in the normal direction Ds. The six-layer through substrate is also called a core layer. A layer consisting of one insulating layer **101** and one wiring layer **102** formed on the one side and the other side of the core layer in the normal direction Ds is also called a build layer. The printed circuit board **100** of the present embodiment is the ten-layer build-up substrate in which two build layers are disposed on both of one side and the other side of the core layer, which is made up of six layers, in the normal direction Ds. The insulating layers **101** are made of an insulating material, for example, epoxy resin. The wiring layers **102** are made of a conductive material, for example, copper.

[0043] In the printed circuit board **100** of the present embodiment, the first receiver coil **120** and the second receiver coil **130** are formed in the wiring layers **102** excluding predetermined wiring layers **102** out of the ten wiring layers **102**. As shown in FIG. **6**, the first receiver coil **120** and the second receiver coil **130** are disposed inside the transmitter coil **110**. Furthermore, in the printed circuit board **100** of the present embodiment, vias **140** that connect the wiring layers **102** are formed. The transmitter coil **110**, the first receiver coil **120** and the second receiver coil **130** formed in the wiring layers **102** are appropriately connected through the vias **140**. The vias **140** function as conductive penetrating portions that electrically connect the ten wiring layers **102**. The shapes and connections of the first receiver coil **120** and the second receiver coil **130** formed in the wiring layers **102** will be described in detail later. The printed circuit board **100** has multiple pad portions (not shown). As shown in FIG. **5**, one end portion of each terminal **400** having a rod shape is connected to the printed circuit board **100** to be connected to the pad portion.

[0044] As the terminals **400**, for example, three terminals for power supply, ground, and output are provided. For example, the terminal **400** for output is connected to the ECU **4** and is used to output the rotation angle of the detection object to the ECU **4**. The number of terminals **400** is not particularly limited, and the connection destinations can be changed appropriately depending on the number of terminals **400**.

[0045] The circuit board **200** is disposed via a joining member (not shown) on a portion of the printed circuit board **100** which is different from portions where the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130** are formed. The circuit board **200** is connected to the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130** via the connection wire **150** provided in the printed circuit board **100**.

[0046] The circuit board **200** includes a microcomputer including a CPU and storage units such as a ROM, a RAM, and a non-volatile RAM, or the like, and is connected to the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130**. The circuit board **200** realizes various control operations by the CPU reading and executing programs from the ROM or the non-volatile RAM. The ROM or the non-volatile RAM stores in advance various data (for example, initial values, look-up tables, maps, and the like) used when the programs are executed. A storage medium such as the ROM is a non-transient tangible storage medium. The CPU is an abbreviation for a Central Processing Unit, the ROM is an abbreviation for a Read Only Memory, and the RAM is an abbreviation for a Random Access Memory.

[0047] Specifically, as shown in FIG. **7**, the circuit board **200** includes a signal processing unit **210** connected to the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130** to perform predetermined processing. The signal processing unit **210** includes, for example, an oscillation unit **220**, a demodulation unit **230**, an AD conversion unit **240**, an angle calculation unit **250**, an output unit **260**, and a power supply unit **300**. In the following, a typical example will be described in which the signal is converted into a digital signal for processing. Meanwhile, in the case of processing with an analog signal, the signal processing unit **210** does not need to include the AD conversion unit **240** and the like.

[0048] As shown in FIGS. **4** and **5**, the sealing member **500** integrally seals the printed circuit board **100**, the circuit board **200**, and the terminals **400**, so that one end portion connected to the printed circuit board **100** in the terminals **400** and the other end portion on an opposite side are exposed. In the following, a portion of the sealing member **500** that is shaped as the arc plate along the shape of the printed circuit board **100** will be referred to as a main portion **510**, and a portion that seals the terminals **400** and connects to an external connector will be referred to as a connector portion **520**. The main portion **510** is formed along the shape of the printed circuit board **100**, and at least an inner edge portion is formed to coincide an arc of a virtual circle formed around the shaft **10**. The connector portion **520** is, for example, in the shape of a generally rectangular tube extending along the normal direction, and has an opening **520***a* that exposes the other end portions of the terminals **400** on the side opposite the main portion **510**. The sealing member **500** is made of, for example, a thermosetting resin or a thermoplastic resin.

[0049] In the sealing member **500**, a collar portion **530** into which a fastening member for being fixed to the fixing base **40** is inserted is provided in a stepped portion in both end portions in the circumferential direction of the main portion **510** having the arc plate shape. The collar portion **530** is configured by disposing a metallic collar **532** in a through hole **531** penetrating through the main portion **510** in the thickness direction. It is not necessary to provide the stepped portion in both end portions in the circumferential direction of the main portion **510**, and shapes of both end portions of the main portion **510** can be appropriately changed to match a shape of side to be fixed.

[0050] The above-described configuration is the configuration of the position detection device S**1** in the present embodiment. As shown in FIG. **2**, the position detection device S**1** is disposed in the fixing base **40** to face the rotating flat plate **30** in the axial direction Da. Specifically, as shown in FIGS. **2** and **3**, the position detection device S**1** is disposed as follows. When the rotating flat plate **30** is rotated, a state where the respective coils **110**, **120**, and **130** and the protrusion portion **32** of the rotating flat plate **30** face each other in the axial direction Da and a state where both of these do not face each other are alternately repeated.

[Signal Processing Unit]

[0051] Next, an operation of the signal processing unit **210** in the above-described circuit board **200** will be described.

[0052] As shown in FIG. **7**, the oscillation unit **220** is connected to both ends of the transmitter coil **110**, and applies an alternating current of a predetermined frequency. For example, two capacitors **161** and **162** are connected in series between both ends of the transmitter coil **110** and the oscillation unit **220**, and a portion connecting the capacitors **161** and **162** to each other is connected to ground. The transmitter coil **110** generates the magnetic field in the axial direction Da which passes through a region surrounded by the first receiver coil **120** and a region surrounded by the second receiver coil **130**. However, a method for connecting the transmitter coil **110** and the oscillation unit **220** can be appropriately changed. For example, one capacitor may be disposed between both ends of the transmitter coil **110** and the oscillation unit **220**.

[0053] The demodulation unit **230** is connected to both ends of the first receiver coil **120** and both ends of the second receiver coil **130**. Then, the demodulation unit **230** generates a first demodulation signal by demodulating a first voltage value V**1** of the first receiver coil **120**, which will be described later, and generates a second demodulation signal by demodulating a second voltage value V**2** of the second receiver coil **130**, which will be described later.

[0054] The AD conversion unit **240** is connected to, for example, the demodulation unit **230** and the angle calculation unit **250**. The AD conversion unit **240** outputs a first conversion signal S obtained by performing AD conversion on the first demodulation signal and a second conversion signal C obtained by performing AD conversion on the second demodulation signal, to the angle calculation unit **250**.

[0055] The angle calculation unit **250** calculates the rotation angle of the rotating flat plate **30** by, for example, calculating an arctangent function using the first conversion signal S and the second conversion signal C.

[0056] The output unit **260** outputs, for example, the rotation angle of the rotating flat plate **30** obtained by the calculation in the angle calculation unit **250** to the terminal **400** for output.

[0057] The power supply unit **300** is connected to each of the units **220** to **260** of the signal

processing unit **210**, and supplies power to each of the units **220** to **260**.
[0058] The basic configuration of the signal processing unit **210** has been described above.
[0059] Next, the first voltage value V**1** of the first receiver coil **120** and the second voltage value V**2** of the second receiver coil **130** when the rotating flat plate **30** is rotated will be described.
[0060] First, an alternating current of a predetermined frequency is applied from the oscillation unit **220** to the transmitter coil **110**. Accordingly, electromagnetic induction is generated in the transmitter coil **110**. Then, the generated electromagnetic induction causes inductive coupling between the transmitter coil **110** and the first and second receiver coils **120** and **130**. Then, a magnetic field in the axial direction Da which passes through a region surrounded by the first receiver coil **120** and a region surrounded by the second receiver coil **130** is generated. In addition, since the magnetic field changes due to the alternating current, the first voltage value V**1**, which is an induced electromotive force generated in the first receiver coil **120**, and the second voltage value V**2**, which is an induced electromotive force generated in the second receiver coil **130**, change due to electromagnetic induction.
[0061] When the protrusion portion **32** of the rotating flat plate **30** faces the transmitter coil **110**, the first receiver coil **120**, and the second receiver coil **130**, electromagnetic induction generates an eddy current, which is an induced current, in the protrusion portion **32**, and a magnetic field is generated due to the eddy current. Therefore, in the magnetic fields in the axial direction Da which pass through the region surrounded by the first receiver coil **120** and the region surrounded by the second receiver coil **130**, the magnetic field which passes through a portion facing the protrusion portion **32** is offset by the magnetic field caused by the eddy current. As a result, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** change.
[0062] As described above, the plurality of protrusion portions **32** are arranged at an interval in the circumferential direction, and the depression portion **31** is provided between the adjacent protrusion portions **32**. In this manner, an area facing the protrusion portion **32** is changed in conjunction with the rotation of the rotating flat plate **30**, and a size of the portion facing the protrusion portion **32** in the magnetic fields in the axial direction Da which pass through the region surrounded by the first receiver coil **120** and the region surrounded by the second receiver coil **130** is periodically changed. Therefore, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** in conjunction with the rotation of the rotating flat plate **30** are periodically changed. In this manner, the rotating flat plate **30** of the present embodiment changes the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** according to a rotation position of the rotating flat plate **30**.
[Details of the First Receiver Coil and the Second Receiver Coil]
[0063] Next, the shapes and connections of the first receiver coil **120** and the second receiver coil **130** formed in the wiring layers **102** will be described in detail. As described above, the printed circuit board **100** of the present embodiment is composed of the ten-layer build-up substrate. The first receiver coil **120** and the second receiver coil **130** are formed in the wiring layers **102** excluding predetermined wiring layers **102** among the ten wiring layers **102**.
[0064] As shown in FIG. **6**, the first receiver coil **120** and the second receiver coil **130** of the present embodiment are primary in a spiral shape, as shown in FIG. **6**. The first receiver coil **120** includes a first forward spiral portion **121** and a first reverse spiral portion **122**. The first forward spiral portion **121** and the first reverse spiral portion **122** are portions of the first receiver coil **120** that form spiral shapes. Furthermore, the second receiver coil **130** of the present embodiment includes a second forward spiral portion **131** and a second reverse spiral portion **132**. The second forward spiral portion **131** and the second reverse spiral portion **132** are portions of the second receiver coil **130** that form spiral shapes.
[0065] The first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward

spiral portion **131** and the second reverse spiral portion **132** are formed side by side at a predetermined interval along the circumferential direction of the printed circuit board **100**, which extends in the shape of the arc plate. Specifically, the first forward spiral portion **121**, the second forward spiral portion **131**, the first reverse spiral portion **122** and the second reverse spiral portion **132** are formed in the stated order from one side to the other side in the circumferential direction of the printed circuit board **100**.

[0066] Each of the first forward spiral portion **121** and the first reverse spiral portion **122** has a spiral pattern shape formed so as to draw a rectangle while changing a diameter. In each of the first forward spiral portion **121** and the first reverse spiral portion **122**, a coil is wound three turns in the same direction in each wiring layer **102**, excluding two central wiring layers **102**, among the ten wiring layers **102**. However, winding directions (that is, spiral directions) of the coils of the first forward spiral portion **121** and the first reverse spiral portion **122** are opposite to each other. For example, the winding direction of the coil of the first forward spiral portion **121** is clockwise as viewed from the one side of the normal direction Ds. In contrast, the winding direction of the coil of the first reverse spiral portion **122** is counterclockwise as viewed from the one side of the normal direction Ds.

[0067] Similarly to the first forward spiral portion **121** and the first reverse spiral portion **122**, each of the second forward spiral portion **131** and the second reverse spiral portion **132** has a spiral pattern shape formed so as to draw a rectangle while changing a diameter. In addition, in each of the second forward spiral portion **131** and the second reverse spiral portion **132**, similarly to the first forward spiral portion **121** and the first reverse spiral portion **122**, a coil is wound three turns in each wiring layer **102**, excluding the two central wiring layers **102**, among the ten wiring layers **102**. However, winding directions of the second forward spiral portion **131** and the second reverse spiral portion **132** are opposite to each other. For example, the coil of the second forward spiral portion **131** is wound clockwise. In contrast, the coil of the second reverse spiral portion **132** is wound counterclockwise.

[0068] Moreover, the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** are formed in a plurality of wiring layers **102** among the wiring layers **102** of the ten-layer build-up substrate. The first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** are formed such that the spiral portions formed in the respective wiring layers **102** overlap in the normal direction Ds.

[0069] For example, the first forward spiral portion **121** formed in the plurality of wiring layers **102** is formed such that the first forward spiral portion **121** formed in the respective wiring layers **102** overlaps with each other in the normal direction Ds. Furthermore, the first reverse spiral portion **122** formed in the plurality of wiring layers **102** is formed such that the first reverse spiral portion **122** formed in the respective wiring layers **102** overlaps with each other in the normal direction Ds. The second forward spiral portion **131** formed in the plurality of wiring layers **102** is formed such that the second forward spiral portion **131** formed in the respective wiring layers **102** overlaps with each other in the normal direction Ds. Furthermore, the second reverse spiral portion **132** formed in the plurality of wiring layers **102** is formed such that the second reverse spiral portion **132** formed in the respective wiring layers **102** overlaps with each other in the normal direction Ds.

[0070] Here, as shown in FIG. **8**, the ten wiring layers **102** are referred to as a first wiring layer **1001**, a second wiring layer **1002**, a third wiring layer **1003**, a fourth wiring layer **1004**, a fifth wiring layer **1005**, a sixth wiring layer **1006**, a seventh wiring layer **1007**, an eighth wiring layer **1008**, a ninth wiring layer **1009**, and a tenth wiring layer **1010** from the one side to the other side in the normal direction Ds.

[0071] In the present embodiment, the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** are formed in the same wiring layers **102** among the ten wiring layers **102**. Specifically, the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** are formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010**. In other words, the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** are formed in the wiring layers **102**, excluding the fifth wiring layer **1005** and the sixth wiring layer **1006**, among the ten wiring layers **102**.

[0072] Each of the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** is formed by winding the coil three turns in each of the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010**. Therefore, the total number of turns of the coil of each of the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** in the present embodiment is twenty four.

[0073] Each of the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** is connected by vias **140** formed in the printed circuit board **100**.

[0074] Here, a connection method of the first forward spiral portion **121** formed in each of the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** is similar to a connection method of each of the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132**. For this reason, in the present embodiment, only the connection method of the first forward spiral portion **121** will be described in detail with reference to FIG. **9** and FIG. **10**, and detailed descriptions of the connection methods of the first reverse spiral portion **122**, the second forward spiral portion **131** and the second reverse spiral portion **132** will be omitted.

[0075] First, the vias **140** that connect the first forward spiral portion **121** will be described. As shown in FIG. **9**, in the printed circuit board **100** of the present embodiment, seven vias **140** are formed to connect the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the fifth wiring layer **1005**, the sixth wiring layer **1006**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010**. Specifically, in the printed circuit board **100**, a first via **1401** that connects the first wiring layer **1001** and the second wiring layer **1002**, and a second via **1402** that connects the second wiring layer **1002** and the third wiring layer **1003** are formed. In addition, in the printed circuit board **100**, a third via **1403**, a fourth via **1404**, and a fifth via **1405** that connect the third wiring layer **1003**, the fourth wiring layer **1004**, the fifth wiring layer **1005**, and the sixth wiring layer **1006** are formed. Furthermore, in the printed circuit board **100**, a sixth via **1406** that connects the eighth wiring layer **1008** and the ninth wiring layer **1009**, and a seventh via **1407** that connects the ninth wiring layer **1009** and the tenth wiring layer **1010** are formed.

[0076] The first via **1401** is formed to penetrate one insulating layer **101** between the first wiring layer **1001** and the second wiring layer **1002** in the normal direction Ds. The second via **1402** is formed to penetrate one insulating layer **101** between the second wiring layer **1002** and the third wiring layer **1003** in the normal direction Ds. The third via **1403**, the fourth via **1404** and the fifth via **1405** are formed to penetrate the five insulating layers **101** between the third wiring layer **1003** and the eighth wiring layer **1008** in the normal direction Ds. The sixth via **1406** is formed to penetrate one insulating layer **101** between the eighth wiring layer **1008** and the ninth wiring layer

1009 in the normal direction Ds. The seventh via **1407** is formed to penetrate one insulating layer **101** between the ninth wiring layer **1009** and the tenth wiring layer **1010** in the normal direction Ds.

[0077] That is, the vias **140** include the first via **1401**, the second via **1402**, the sixth via **1406** and the seventh via **1407** that are formed to penetrate one insulating layer **101**. Furthermore, the vias **140** include the third via **1403**, the fourth via **1404** and the fifth via **1405** that are formed to penetrate the five insulating layers **101**. Each of the third via **1403**, the fourth via **1404** and the fifth via **1405** serves as a first conductive portion formed to penetrate five insulating layers **101**. Each of the first via **1401**, the second via **1402**, the sixth via **1406** and the seventh via **1407** serves as a second conductive portion formed to penetrate one insulating layer **101**.

[0078] The first via **1401**, the second via **1402**, the sixth via **1406** and the seventh via **1407** are formed by copper plating holes created in the insulating layer **101** by a laser, for example, with respect to the build layer. The third via **1403**, the fourth via **1404**, and the fifth via **1405** are formed, for example, by copper plating in through holes formed through all of the wiring layers **102** and insulating layers **101** in the core layer composed of the six-layer through substrate. In FIG. **9**, the insulating layers **101** between the wiring layers **102** are omitted for ease of viewing.

[0079] As shown in FIG. **6** and FIG. **9**, the first via **1401** and the seventh via **1407** are formed at positions overlapping each other in the normal direction Ds. The first via **1401** and the seventh via **1407** are formed at positions overlapping with the second via **1402**, the third via **1403**, the fourth via **1404**, the fifth via **1405**, and the sixth via **1406** in the normal direction Ds.

[0080] The second via **1402** and the sixth via **1406** are formed at positions where they overlap each other in the normal direction Ds. The second via **1402** and the sixth via **1406** are formed at positions not overlapping with the third via **1403**, the fourth via **1404**, and the fifth via **1405** in the normal direction Ds.

[0081] The third via **1403**, the fourth via **1404**, and the fifth via **1405** are formed at positions not overlapping with each other in the normal direction Ds.

[0082] As shown in FIG. **6**, the first via **1401**, the second via **1402**, the third via **1403**, the fourth via **1404**, the fifth via **1405**, the sixth via **1406** and the seventh via **1407** are formed at positions not overlapping with the first forward spiral portion **121** in the normal direction Ds. Specifically, the first via **1401**, the third via **1403**, the fifth via **1405** and the seventh via **1407** are formed inside the spiral shape of the first forward spiral portion **121**. In contrast, the second via **1402**, the fourth via **1404** and the sixth via **1406** are formed outside the spiral shape of the first forward spiral portion **121**. In other words, the first via **1401** to the seventh via **1407** are arranged alternately inside and outside the first forward spiral portion **121**, with the vias **1401**, **1403**, **1405**, and **1407** formed on the inside of the first forward spiral portion **121** and the vias **1402**, **1404**, and **1406** formed on the outside of the first forward spiral portion **121**.

[0083] The first via **1401** in the present embodiment corresponds to an inside first-layer-to-second-layer conductive portion. The second via **1402** corresponds to an outside second-layer-to-third-layer conductive portion. The third via **1403** corresponds to an inside third-layer-to-fourth-layer conductive portion. The fourth via **1404** corresponds to an outside fourth-layer-to-seventh-layer conductive portion. The fifth via **1405** corresponds to an inside seventh-layer-to-eighth-layer conductive portion. The sixth via **1406** corresponds to an outside eighth-layer-to-ninth-layer conductive portion. The seventh via **1407** corresponds to an inside ninth-layer-to-tenth-layer conductive portion.

[0084] Next, a method of connecting the first forward spiral portion **121** formed in each of the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** will be described. In the following description, the first forward spiral portion **121** formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** will also be referred to as a first spiral part **1201**, a second spiral part **1202**, a third spiral part **1203**, a fourth spiral part **1204**, a seventh spiral part **1207**, an eighth spiral part **1208**, a ninth spiral part **1209** and a tenth spiral part **1210**, respectively.

[0085] In the first forward spiral portion **121**, the parts formed in the respective wiring layer **102** are electrically connected by the first via **1401**, the second via **1402**, the third via **1403**, the fourth via **1404**, the fifth via **1405**, the sixth via **1406** and the seventh via **1407**. Specifically, as shown in FIG. **9**, the first spiral part **1201** and the second spiral part **1202** are connected through the first via **1401**. The second spiral part **1202** and the third spiral part **1203** are connected through the second via **1402**. The third spiral part **1203** and the fourth spiral part **1204** are connected through the third via **1403**. The fourth spiral part **1204** and the seventh spiral part **1207** are connected through the fourth via **1404**. The seventh spiral part **1207** and the eighth spiral part **1208** are connected through the fifth via **1405**. The eighth spiral part **1208** and the ninth spiral part **1209** are connected through the sixth via **1406**. The ninth spiral part **1209** and the tenth spiral part **1210** are connected through the seventh via **1407**.

[0086] As described above, in the present embodiment, the first spiral part **1201**, the second spiral part **1202**, the third spiral part **1203**, the fourth spiral part **1204**, the seventh spiral part **1207**, the eighth spiral part **1208**, the ninth spiral part **1209** and the tenth spiral portion **1210** are electrically connected through the first via **1401**, the second via **1402**, the third via **1403**, the fourth via **1404**, the fifth via **1405**, the sixth via **1406** and the seventh via **1407**.

[0087] The first spiral part **1201** and the second spiral part **1202** are connected by the first via **1401** formed inside the first forward spiral portion **121**. The second spiral part **1202** and the third spiral part **1203** are connected by the second via **1402** formed outside the first forward spiral portion **121**. The third spiral part **1203** and the fourth spiral part **1204** are connected by the third via **1403** formed inside the first forward spiral portion **121**. The fourth spiral part **1204** and the seventh spiral part **1207** are connected by the fourth via **1404** formed outside the first forward spiral portion **121**. The seventh spiral part **1207** and the eighth spiral part **1208** are connected by the fifth via **1405** formed inside the first forward spiral portion **121**. The eighth spiral part **1208** and the ninth spiral part **1209** are connected by the sixth via **1406** formed outside the first forward spiral portion **121**. The ninth spiral part **1209** and the tenth spiral part **1210** are connected by the seventh via **1407** formed inside the first forward spiral portion **121**.

[0088] In this manner, the first forward spiral portion **121** formed in each wiring layer **102** is electrically connected by the first via **1401** to the seventh via **1407** formed alternately inside and outside the first forward spiral portion **121**.

[0089] In addition, the first reverse spiral portion **122** formed in each wiring layer **102** is electrically connected by the first via **1401** to the seventh via **1407** formed alternately on the inside and outside of the first reverse spiral portion **122**. The second forward spiral portion **131** formed in each wiring layer **102** is electrically connected by a first via **1401** to a seventh via **1407** formed alternately inside and outside the second forward spiral portion **131**. The second reverse spiral portion **132** formed in each wiring layer **102** is electrically connected by a first via **1401** to a seventh via **1407** formed alternately inside and outside the second reverse spiral portion **132**.

[0090] The reason why the first via **1401** to the seventh via **1407** are alternately formed inside and outside each of the first forward spiral portion **121**, the first reverse spiral portion **122**, the second forward spiral portion **131**, and the second reverse spiral portion **132** will be explained using the first forward spiral portion **121**. As described above, the first spiral part **1201**, the second spiral part **1202**, the third spiral part **1203**, the fourth spiral part **1204**, the seventh spiral part **1207**, the eighth spiral part **1208**, the ninth spiral part **1209** and the tenth spiral part **1210** have coils wound in the same direction (clockwise in the present embodiment). In addition, the first spiral part **1201**, the second spiral part **1202**, the third spiral part **1203**, the fourth spiral part **1204**, the seventh spiral part **1207**, the eighth spiral part **1208**, the ninth spiral part **1209** and the tenth spiral part **1210** are

formed to overlap in the normal direction Ds.
[0091] Therefore, among the first spiral part **1201**, the second spiral part **1202**, the third spiral part **1203**, the fourth spiral part **1204**, the seventh spiral part **1207**, the eighth spiral part **1208**, the ninth spiral part **1209** and the tenth spiral part **1210**, adjacent spiral parts are configured by alternately arranging a part wound clockwise from the outside toward the inside and a part would clockwise from the inside toward the outside.
[0092] For example, the first spiral part **1201** is formed by winding a coil from the outside toward the inside. In this case, the second spiral part **1202**, which is formed in the second wiring layer **1002** adjacent to the first wiring layer **1001** in which the first spiral part **1201** is formed, is formed by winding a coil from the inside to the outside. The third spiral part **1203**, which is formed in the third wiring layer **1003** adjacent to the second wiring layer **1002** in which the second spiral part **1202** is formed, is formed by winding a coil from the outside to the inside.
[0093] In the present embodiment, the first forward spiral portion **121** is formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010**, excluding the fifth wiring layer **1005** and the sixth wiring layer **1006**, among the first wiring layer **1001** to the tenth wiring layer **1010**.
[0094] In FIG. **9**, each of the wiring layers **102**, the first forward spiral portion **121**, and the vias **140** are schematically shown to clarify the electrical connections between each of the wiring layers **102** and the first forward spiral portion **121**. The wiring layers **102** in which the first forward spiral portion **121** is formed are indicated by solid lines, and the wiring layers **102** in which the first forward spiral portion **121** is not formed are indicated by dashed lines. Portions where the first spiral part **1201**, the second spiral part **1202**, the third spiral part **1203**, the fourth spiral part **1204**, the seventh spiral part **1207**, the eighth spiral part **1208**, the ninth spiral part **1209** and the tenth spiral part **1210** are electrically connected are indicated by thick lines.
[0095] In addition, in FIG. **10**, in each of the wiring layers **102** in which the first forward spiral portion **121** is formed, the vias **140** that appear when viewed from the one side in the normal direction Ds are indicated by solid lines, and the vias **140** that do not appear are indicated by dashed lines. Moreover, the electrical connections between the vias **140** are indicated by dash-dotted lines.
[0096] Next, the reason why the first forward spiral portion **121** is not formed in the fifth wiring layer **1005** and the sixth wiring layer **1006** among the first wiring layer **1001** to the tenth wiring layer **1010** will be described.
[0097] If the first forward spiral portion **121** is formed in the fifth wiring layer **1005** and the sixth wiring layer **1006**, it is necessary to add vias **140** to the printed circuit board **100** to connect the first forward spiral section **121** formed in the fifth wiring layer **1005** and the sixth wiring layer **1006**.
[0098] Specifically, when forming the first forward spiral portion **121** in the fifth wiring layer **1005** and the sixth wiring layer **1006**, the printed circuit board **100** requires a via **140** to connect the fourth spiral part **1204** and the first forward spiral portion **121** formed in the fifth wiring layer **1005**. In addition, the printed circuit board **100** requires a via **140** to connect the first forward spiral portion **121** formed in the fifth wiring layer **1005** and the first forward spiral portion **121** formed in the sixth wiring layer **1006**. Furthermore, the printed circuit board **100** requires a via **140** to connect the first forward spiral portion **121** formed in sixth wiring layer **1006** and the seventh spiral part **1207**.
[0099] Here, a board in which the first forward spiral portion **121** is formed in the fifth wiring layer **1005** and the sixth wiring layer **1006** shown in FIG. **11** is referred to as comparative printed board **100**A. In the comparative printed circuit board **100**A, the first forward spiral portion **121** formed in the fifth wiring layer **1005** is referred to as a fifth spiral part **1205**, and the first forward spiral portion **121** formed in the sixth wiring layer **1006** is referred to as a sixth spiral part **1206**. A via

140 for connecting the fourth spiral part **1204** and the fifth spiral part **1205** is referred to as an eighth via **1408**, and a via **140** for connecting the fifth spiral part **1205** and the sixth spiral part **1206** is referred to as a ninth via **1409**. In a case where the eighth via **1408** and the ninth via **1409** are formed, the fourth via **1404** connects the sixth spiral part **1206** and the seventh spiral part **1207**.
[0100] As shown in FIG. **11**, the eighth via **1408** connecting the fourth spiral part **1204** and the fifth spiral part **1205** is formed, similarly to the third via **1403**, the fourth via **1404** and the fifth via **1405**, to penetrate the core layer composed of the six-layer through substrate. In addition, the ninth via **1409** connecting the fifth spiral part **1205** and the sixth spiral part **1206** is also formed, similarly to the third via **1403**, the fourth via **1404** and the fifth via **1405**, to penetrate the core layer composed of the six-layer through substrate.
[0101] The first spiral part **1201** to the tenth spiral part **1210** need to be formed so as to overlap in the normal direction Ds. Then, the first via **1401** to the ninth via **1409** need to be formed at positions not overlapping with the first forward spiral portion **121** in the normal direction Ds, as shown in FIG. **12**.
[0102] Furthermore, the first via **1401** to the ninth via **1409** need to be formed alternately inside and outside the first forward spiral portion **121**, as described above. Therefore, the eighth via **1408**, which is next to the third via **1403** formed inside the first forward spiral portion **121**, is formed outside the first forward spiral portion **121** as shown in FIG. **12**. Furthermore, the ninth via **1409**, which is next to the eighth via **1408** formed outside the first forward spiral portion **121**, is formed inside the first forward spiral portion **121**.
[0103] Furthermore, when forming the first forward spiral portion **121** in all of the ten wiring layers **102** of the comparative printed circuit board **100**A, the first forward spiral portion **121** needs to be formed in a position not overlapping with the first via **1401** to the ninth via **1409** in the normal direction Ds. This limits the surface area of the wiring layer **102** on which the first forward spiral portion **121** can be formed. Specifically, by adding the ninth via **1409** to the comparative printed circuit board **100**A, the surface area available for forming a spiral coil inside the first forward spiral portion **121** in the wiring layer **102** is reduced. In other words, compared to the printed circuit board **100** of the present embodiment, the comparative printed circuit board **100**A has a reduced surface area available for forming the first forward spiral portion **121** in the wiring layer **102**.
[0104] As a result, by adding the ninth via **1409** inside the first forward spiral section **121**, the comparative printed circuit board **100**A has a reduced number of turns of the coil that can be formed inside the first forward spiral portion **121** compared to the printed circuit board **100** of the present embodiment. For example, by adding the ninth via **1409**, the possible number of turns of the coil in the first forward spiral portion **121** is reduced from three to two. As a result, the first forward spiral portion **121** in the comparative printed circuit board **100**A is configured by the coil wound twenty turns. In other words, the total number of turns of the coil that constitutes the first forward spiral portion **121** is twenty.
[0105] The first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** with the rotation of the rotating flat plate **30** change according to the total number of turns of each of the coils of the first receiver coil **120** and the second receiver coil **130**. Specifically, the first voltage value V**1** and the second voltage value V**2** increase with increase in the total number of turns of the coils of the first receiver coil **120** and the second receiver coil **130**.
[0106] As a result, the comparative printed circuit board **100**A, in which the total number of turns of the coil that constitutes the first forward spiral portion **121** is reduced compared to the printed circuit board **100** of the present embodiment, has smaller first voltage value V**1** and second voltage value V**2** compared to the printed circuit board **100** of the present embodiment.
[0107] In contrast, the position detection device S**1** of the present embodiment includes the printed circuit board **100**, which is the multilayer substrate with the ten wiring layers **102** and the insulating

layers **101** alternately laminated between each of the ten wiring layers **102**. The ten wiring layers **102** are electrically connected by the vias **140** formed to penetrate at least one insulating layer **101**.

[0108] The first receiver coil **120** and the second receiver coil **130** are wound three turns in each of the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009**, and the tenth wiring layer **1010**, excluding the fifth wiring layer **1005** and the sixth wiring layer **1006**, among the ten wiring layers **102**. The first receiver coil **120** and the second receiver coil **130** formed in each of the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** are connected through the vias **140**.

[0109] According to this configuration, it is possible to reduce the number of the vias **140** for connecting the first receiver coil **120** and the second receiver coil **130** compared to a case where the first receiver coil **120** and the second receiver coil **130** are formed in all of the ten wiring layers **102**. Thus, in each wiring layer **102** in which the first receiver coil **120** and the second receiver coil **130** are formed, the surface area available for forming the first receiver coil **120** and the second receiver coil **130** can be increased.

[0110] Therefore, it is easier to increase the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130** in each of the wiring layers **102**. Furthermore, by increasing the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130**, the total number of turns of each of the coils of the first receiver coil **120** and the second receiver coil **130** in the entire printed circuit board **100** can be increased.

[0111] Furthermore, by increasing the total number of turns of the coils of each of the first receiver coil **120** and the second receiver coil **130**, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** can be increased.

Second Embodiment

[0112] Next, a second embodiment will be described with reference to FIG. **13** and FIG. **14**. In the present embodiment, the configuration of a printed circuit board **100** is different from that of the first embodiment. The other configurations are the same as those of the first embodiment. Therefore, in the present embodiment, portions different from the first embodiment will be mainly described, and description of portions similar to the first embodiment may be omitted.

[0113] The printed circuit board **100** of the present embodiment includes a four-layer through substrate in which insulating layers **101** and wiring layers **102** are alternately laminated. The printed circuit board **100** is composed of an eight-layer build-up substrate in which two insulating layers **101** and two wiring layers **102** are further laminated on both of one side and the other side of the through substrate in the normal direction Ds. The printed circuit board **100** of the present embodiment is an eight-layer build-up board in which two build layers are disposed on both of one side and the other side of a core layer, which is made up of four layers, in the normal direction Ds. As shown in FIG. **13**, the printed circuit board **100** of the present embodiment includes a first wiring layer **1001** to an eighth wiring layer **1008**.

[0114] Furthermore, in each of the eight wiring layers **102** of the printed circuit board **100** excluding two central wiring layers **102**, the first receiver coil **120** and the second receiver coil **130** are wound three turns each in the same direction. That is, the first receiver coil **120** and the second receiver coil **130** are formed by being wound three turns each in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the sixth wiring layer **1006**, the seventh wiring layer **1007** and the eighth wiring layer **1008**, excluding the fourth wiring layer **1004** and the fifth wiring layer **1005**.

[0115] In addition, in the printed circuit board **100**, a first via **1401** that connects the first wiring layer **1001** and the second wiring layer **1002**, and a second via **1402** that connects the second wiring layer **1002** and the third wiring layer **1003** are formed. Furthermore, in the printed circuit board **100**, a third via **1403** that connects the third wiring layer **1003**, the fourth wiring layer **1004**, the fifth wiring layer **1005** and the sixth wiring layer **1006** is formed. Furthermore, in the printed circuit board **100**, a fourth via **1404** that connects the sixth wiring layer **1006** and the seventh wiring layer **1007**, and a fifth via **1405** that connects the seventh wiring layer **1007** and the eighth wiring layer **1008** are formed.

[0116] The first via **1401** is formed to penetrate one insulating layer **101** between the first wiring layer **1001** and the second wiring layer **1002** in the normal direction Ds. The second via **1402** is formed to penetrate one insulating layer **101** between the second wiring layer **1002** and the third wiring layer **1003** in the normal direction Ds. The third via **1403** is formed to penetrate three insulating layers **101** between the third wiring layer **1003** and the sixth wiring layer **1006** in the normal direction Ds. The fourth via **1404** is formed to penetrate one insulating layer **101** between the sixth wiring layer **1006** and the seventh wiring layer **1007** in the normal direction Ds. The fifth via **1405** is formed to penetrate one insulating layer **101** between the seventh wiring layer **1007** and the eighth wiring layer **1008** in the normal direction Ds.

[0117] That is, the vias **140** include the first via **1401**, the second via **1402**, the fourth via **1404** and the fifth via **1405** formed to penetrate one insulating layer **101**, and the third via **1403** formed to penetrate three insulating layers **101**. The third via **1403** serves as a first conductive portion formed by penetrating three insulating layers **101**. The first via **1401**, the second via **1402**, the fourth via **1404** and the fifth via **1405** function as a second conductive portion formed by penetrating one insulating layer **101**.

[0118] As shown in FIG. **13** and FIG. **14**, the first via **1401** and the fifth via **1405** are formed at positions overlapping each other in the normal direction Ds. The first via **1401** and the fifth via **1405** are formed at positions not overlapping with the second via **1402**, the third via **1403**, and the fourth via **1404** in the normal direction Ds.

[0119] The second via **1402** is formed at a position not overlapping with the third via **1403** and the fourth via **1404** in the normal direction Ds.

[0120] The third via **1403** and the fourth via **1404** are formed at positions not overlapping with each other in the normal direction Ds.

[0121] The first via **1401**, the second via **1402**, the third via **1403**, the fourth via **1404** and the fifth via **1405** are formed at positions not overlapping with the first receiver coil **120** and the second receiver coil **130** in the normal direction Ds, as shown in FIG. **14**. Specifically, the first via **1401**, the third via **1403** and the fifth via **1405** are formed inside the spiral shapes of the first receiver coil **120** and the second receiver coil **130**. In contrast, the second via **1402** and the fourth via **1404** are formed outside the spiral shapes of the first receiver coil **120** and the second receiver coil **130**. The first via **1401** to the fifth via **1405** are arranged alternately inside and outside each of the first receiver coil **120** and the second receiver coil **130**, with the vias **1401**, **1403**, and **1405** formed inside and the vias **1402** and **1404** formed outside.

[0122] The first receiver coil **120** formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the sixth wiring layer **1006**, the seventh wiring layer **1007** and the eighth wiring layer **1008** is electrically connected by the first via **1401** to the fifth via **1405** formed alternately inside and outside the first receiver coil **120**. In addition, the second receiver coil **130** formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the sixth wiring layer **1006**, the seventh wiring layer **1007** and the eighth wiring layer **1008** is electrically connected by the first vias **1401** to the fifth vias **1405** formed alternately inside and outside the second receiver coil **130**.

[0123] For example, the first receiver coil **120** formed in the first wiring layer **1001** and the first receiver coil **120** formed in the second wiring layer **1002** are connected by the first via **1401** formed inside the first receiver coil **120**. The first receiver coil **120** formed in the second wiring layer **1002** and the first receiver coil **120** formed in the third wiring layer **1003** are connected by the second via **1402** formed outside the first receiver coil **120**. The first receiver coil **120** formed in the third wiring layer **1003** and the first receiver coil **120** formed in the sixth wiring layer **1006** are connected by the third via **1403** formed inside the first receiver coil **120**. The first receiver coil **120** formed in the sixth wiring layer **1006** and the first receiver coil **120** formed in the seventh wiring layer **1007** are connected by the fourth via **1404** formed outside the first receiver coil **120**. The first receiver coil **120** formed in the seventh wiring layer **1007** and the first receiver coil **120** formed in the eighth wiring layer **1008** are connected by the fifth via **1405** formed inside the first receiver coil **120**.

[0124] According to this configuration, it is possible to reduce the number of the vias **140** for connecting the first receiver coil **120** and the second receiver coil **130** compared to a case where the first receiver coil **120** and the second receiver coil **130** are formed in all of the eight wiring layers **102**. Thus, in each wiring layer **102** in which the first receiver coil **120** and the second receiver coil **130** are formed, the surface area available for forming the first receiver coil **120** and the second receiver coil **130** can be increased.

[0125] Therefore, it is easier to increase the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130** in each of the wiring layers **102**. Furthermore, by increasing the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130**, the total number of turns of each of the coils of the first receiver coil **120** and the second receiver coil **130** in the entire printed circuit board **100** can be increased.

[0126] Furthermore, by increasing the total number of turns of the coils that constitute each of the first receiver coil **120** and the second receiver coil **130**, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** can be increased.

Third Embodiment

[0127] Next, a third embodiment will be described with reference to FIG. **15** and FIG. **16**. In the present embodiment, the configuration of a printed circuit board **100** is different from that of the first embodiment. The other configurations are the same as those of the first embodiment. Therefore, in the present embodiment, portions different from the first embodiment will be mainly described, and description of portions similar to the first embodiment may be omitted.

[0128] The printed circuit board **100** of the present embodiment includes a four-layer through substrate in which insulating layers **101** and wiring layers **102** are alternately laminated. The printed circuit board **100** is composed of a six-layer build-up substrate in which one insulating layer **101** and one wiring layer **102** are further laminated on both of one side and the other side of the through substrate in the normal direction Ds. As shown in FIG. **15**, the printed circuit board **100** of the present embodiment has a first wiring layer **1001** to a sixth wiring layer **1006**.

[0129] Furthermore, in each of the six wiring layers **102** of the printed circuit board **100** excluding two central wiring layers **102**, the first receiver coil **120** and the second receiver coil **130** are wound three turns each in the same direction. That is, the first receiver coil **120** and the second receiver coil **130** are formed by being wound three turns each in the first wiring layer **1001**, the second wiring layer **1002**, the fifth wiring layer **1005** and the sixth wiring layer **1006**, excluding the third wiring layer **1003** and the fourth wiring layer **1004**.

[0130] In addition, in the printed circuit board **100**, a first via **1401** that connects the first wiring layer **1001** and the second wiring layer **1002**, and a second via **1402** that connects the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004** and the fifth wiring layer **1005** are formed. Furthermore, in the printed circuit board **100**, a third via **1403** that connects the fifth wiring layer **1005** and the sixth wiring layer **1006** is formed.

[0131] The first via **1401** is formed to penetrate one insulating layer **101** between the first wiring layer **1001** and the second wiring layer **1002** in the normal direction Ds. The second via **1402** is formed to penetrate three insulating layers **101** between the second wiring layer **1002** and the fifth wiring layer **1005** in the normal direction Ds. The third via **1403** is formed to penetrate one insulating layer **101** between the fifth wiring layer **1005** and the sixth wiring layer **1006** in the normal direction Ds.

[0132] That is, the vias **140** include the first via **1401** and the third via **1403** formed to penetrate one insulating layer **101**, and the second via **1402** formed to penetrate three insulating layers **101**. The second via **1402** serves as a first conductive portion formed to penetrate the three insulating layers **101**. The first via **1401** and the third via **1403** serves as a second conductive portion formed to penetrate one insulating layer **101**.

[0133] The first via **1401**, the second via **1402**, and the third via **1403** are formed at positions not overlapping with each other in the normal direction Ds.

[0134] As shown in FIG. **16**, the first via **1401**, the second via **1402**, and the third via **1403** are formed at positions not overlapping with the first receiver coil **120** and the second receiver coil **130** in the normal direction Ds. Specifically, the first via **1401** and the third via **1403** are formed inside the spiral shapes of the first receiver coil **120** and the second receiver coil **130**. In contrast, the second via **1402** is formed outside the spiral shapes of the first receiver coil **120** and the second receiver coil **130**. The first vias **1401** to the third vias **1403** are arranged alternately inside and outside the first receiver coil **120**, with the vias **1401** and **1403** formed inside and the via **1402** formed outside.

[0135] The first receiver coil **120** and the second receiver coil **130** formed in the first wiring layer **1001**, the second wiring layer **1002**, the fifth wiring layer **1005** and the sixth wiring layer **1006** are electrically connected by the first vias **1401** to the third vias **1403** formed alternately inside and outside the first receiver coil **120**.

[0136] For example, the first receiver coil **120** formed in the first wiring layer **1001** and the first receiver coil **120** formed in the second wiring layer **1002** are connected by the first via **1401** formed inside the first receiver coil **120**. The first receiver coil **120** formed in the second wiring layer **1002** and the first receiver coil **120** formed in the fifth wiring layer **1005** are connected by the second via **1402** formed outside the first receiver coil **120**. The first receiver coil **120** formed in the fifth wiring layer **1005** and the first receiver coil **120** formed in the sixth wiring layer **1006** are connected by the third via **1403** formed inside the first receiver coil **120**.

[0137] According to this configuration, it is possible to reduce the number of the vias **140** for connecting the first receiver coil **120** and the second receiver coil **130** compared to a case where the first receiver coil **120** and the second receiver coil **130** are formed in all of the six wiring layers **102**. Thus, in each wiring layer **102** in which the first receiver coil **120** and the second receiver coil **130** are formed, the surface area available for forming the first receiver coil **120** and the second receiver coil **130** can be increased.

[0138] Therefore, it is easier to increase the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130** in each of the wiring layers **102**. Furthermore, by increasing the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130**, the total number of turns of each of the coils of the first receiver coil **120** and the second receiver coil **130** in the entire printed circuit board **100** can be increased.

[0139] Furthermore, by increasing the total number of turns of the coils that constitute each of the first receiver coil **120** and the second receiver coil **130**, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** can be increased.

Fourth Embodiment

[0140] Next, a fourth embodiment will be described with reference to FIG. **17**. In the present embodiment, pattern shapes of a first receiver coil **120** and a second receiver coil **130** are different from those in the first embodiment. The other configurations are the same as those of the first embodiment. Therefore, in the present embodiment, portions different from the first embodiment will be mainly described, and description of portions similar to the first embodiment may be omitted.

[0141] As shown in FIG. **17**, the first receiver coil **120** and the second receiver coil **130** in the present embodiment are formed into arc frame shapes with the coils wound four turns in the normal direction Ds and with one direction (that is, a circumferential direction of the printed circuit board **100**) as a longitudinal direction.
[0142] The first receiver coil **120** and the second receiver coil **130** are disposed inside the transmitter coil **110** in the normal direction Ds. Moreover, the first receiver coil **120** and the second receiver coil **130** are configured by appropriately connecting different wiring layers through vias **140** so as not to interfere with each other (that is, not to overlap in the same layer).
[0143] The first receiver coil **32** is formed in a pattern shape that draws a sine curve so as to have a sine wave shape in a closed loop. The first receiver coil **120**, which is formed by winding the coil four turns, has each wound portion offset by a predetermined amount in an amplitude direction for each turn. The first receiver coil **120**, which is formed by winding the coil four turns, has each wound portion connected in series and is formed in a single continuous line.
[0144] The second receiver coil **130** is formed in a pattern shape that draws a cosine curve so as to form a cosine wave in a closed loop. The second receiver coil **130**, which is formed by winding the coil four turns, has each wound portion offset by a predetermined amount in an amplitude direction for each turn. The second receiver coil **130**, which is formed by winding the coil four turns, has each wound portion connected in series and is formed in a single continuous line.
[0145] The first receiver coil **120** and the second receiver coil **130** formed by a plurality of turns may be formed with a phase difference. The first receiver coil **32** may be formed in a pattern shape that draws a cosine curve so as to have a cosine wave shape in a closed loop. In this case, the second receiver coil **130** is formed in a pattern shape that draws a sine curve so as to have a sine wave shape in a closed loop.
[0146] The first receiver coil **120** and the second receiver coil **130** are formed in the first wiring layer **1001**, the second wiring layer **1002**, the third wiring layer **1003**, the fourth wiring layer **1004**, the seventh wiring layer **1007**, the eighth wiring layer **1008**, the ninth wiring layer **1009** and the tenth wiring layer **1010** among the ten wiring layers **102**. In other words, the first receiver coil **120** and the second receiver coil **130** are formed in the wiring layers **102**, excluding the fifth wiring layer **1005** and the sixth wiring layer **1006**, among the ten wiring layers **102**.
[0147] Among the ten wiring layers **102**, adjacent wiring layers **102** in which the first receiver coil **120** and the second receiver coil **130** are formed are connected to each other by vias **140**.
[0148] For example, the first receiver coil **120** formed in the first wiring layer **1001** and the first receiver coil **120** formed in the second wiring layer **1002** are connected by a via **140** formed between the first wiring layer **1001** and the second wiring layer **1002**. The first receiver coil **120** formed in the second wiring layer **1002** and the first receiver coil **120** formed in the third wiring layer **1003** are connected by a via **140** formed between the second wiring layer **1002** and the third wiring layer **1003**. The first receiver coil **120** formed in the third wiring layer **1003** and the first receiver coil **120** formed in the fourth wiring layer **1004** are connected by a via **140** formed between the third wiring layer **1003** and the fourth wiring layer **1004**.
[0149] The first receiver coil **120** formed in the fifth wiring layer **1005** and the first receiver coil **120** formed in the seventh wiring layer **1007** are connected by a via **140** formed to penetrate from the fifth wiring layer **1005** to the seventh wiring layer **1007**.
[0150] The first receiver coil **120** formed in the seventh wiring layer **1007** and the first receiver coil **120** formed in the eighth wiring layer **1008** are connected by a via **140** formed between the seventh wiring layer **1007** and the eighth wiring layer **1008**. The first receiver coil **120** formed in the eighth wiring layer **1008** and the first receiver coil **120** formed in the ninth wiring layer **1009** are connected by a via **140** formed between the eighth wiring layer **1008** and the ninth wiring layer **1009**. The first receiver coil **120** formed in the ninth wiring layer **1009** and the first receiver coil **120** formed in the tenth wiring layer **1010** are connected by a via **140** formed between the ninth wiring layer **1009** and the tenth wiring layer **1010**.

[0151] Note that, in FIG. **17**, for clarity, the first receiver coil **120** and the second receiver coil **130**, which are formed in the adjacent wiring layers **102** among the first wiring layer **1001** to the tenth wiring layer **1010**, are indicated using solid and dashed lines.
[0152] According to this configuration, it is possible to reduce the number of the vias **140** for connecting the first receiver coil **120** and the second receiver coil **130** compared to a case where the first receiver coil **120** and the second receiver coil **130** are formed in all of the ten wiring layers **102**. Thus, in each wiring layer **102** in which the first receiver coil **120** and the second receiver coil **130** are formed, the surface area available for forming the first receiver coil **120** and the second receiver coil **130** can be increased.
[0153] Therefore, it is easier to increase the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130** in each of the wiring layers **102**. Furthermore, by increasing the number of turns of the coils that constitute the first receiver coil **120** and the second receiver coil **130**, the total number of turns of each of the coils of the first receiver coil **120** and the second receiver coil **130** in the entire printed circuit board **100** can be increased.
[0154] Furthermore, by increasing the total number of turns of the coils that constitute each of the first receiver coil **120** and the second receiver coil **130**, the first voltage value V**1** generated in the first receiver coil **120** and the second voltage value V**2** generated in the second receiver coil **130** can be increased.
Other Embodiments
[0155] Although the representative embodiments of the present disclosure have been described above, the present disclosure is not limited to the embodiments described above, and can be variously modified as follows, for example.
[0156] In the above embodiment, an example has been described in which the position detection device S**1** is applied to an electrification system mounted on a vehicle, but the present disclosure is not limited to this configuration. For example, the position detection device S**1** may be mounted on something other than a vehicle.
[0157] In the above embodiment, the position detection device S**1** detects the rotation angle of a rotating detection object, but the present disclosure is not limited to this configuration. For example, the position detection device S**1** may detect the amount of displacement of a linearly moving detection body.
[0158] In the above-described embodiments, examples have been described in which the first receiver coil **120** and the second receiver coil **130** are wound two or three turns, but the number of turns is not limited to these examples. For example, the first receiver coil **120** and the second receiver coil **130** may be wound four or more turns.
[0159] In the above-described embodiments, the printed circuit board **100** is composed of the six-layer, eight-layer, or ten-layer build-up substrate, but the number of layers of the printed circuit board **100** is not limited to these examples. The number of layers of the printed circuit board **100** can be changed as appropriate.
[0160] In the above-described embodiments, examples have been described in which the printed circuit board **100** is the six-layer, eight-layer or ten-layer build-up substrate, and the first receiver coil **120** and the second receiver coil **130** are formed in the wiring layers **102** excluding the two central wiring layers **102**, but the present disclosure is not limited to these examples. For example, the first receiver coil **120** and the second receiver coil **130** may be configured not to be formed in a wiring layer **102** different from the two central wiring layers **102** in a six-layer, eight-layer, or ten-layer build-up substrate. Furthermore, the first receiver coil **120** and the second receiver coil **130** may be configured not to be formed in one or three or more wiring layers **102** in a six-layer, eight-layer, or ten-layer build-up substrate.
[0161] In the embodiments described above, it is needless to say that the elements configuring the embodiments are not necessarily essential except in the case where those elements are clearly indicated to be essential in particular, the case where those elements are considered to be obviously

essential in principle, and the like.
[0162] In the embodiments described above, when a numerical value such as the number, a numerical value, an amount, or a range of the constituent elements of the embodiment is referred to, the numerical value is not limited to specific numerical values unless otherwise specified as being essential or obviously limited to the specific numerical values in principle.
[0163] In each of the embodiments, when the shapes, positional relationships, and the like of the constituent elements and the like are referred to, the shapes, positional relationships, and the like are not limited thereto unless otherwise specified or limited to specific shapes, positional relationships, and the like in principle.
[0164] The control unit and the method thereof of the present disclosure may be implemented by a dedicated computer provided by configuring a memory and a processor programmed to execute one or a plurality of functions embodied by a computer program. The control unit and the method therefor of the present disclosure may be realized by a dedicated computer provided by including a processor with one or more dedicated hardware logic circuits. The control unit and the method therefor of the present disclosure may be realized by one or more dedicated computers, each including a combination of a processor and a memory that are programmed to execute one or more functions and a processor formed of one or more hardware logic circuits. The computer programs may be stored, as instructions to be executed by a computer, in a tangible non-transitory computer-readable medium.

Claims

1. A position detection device comprising: a substrate; a transmitter coil formed in the substrate; and a first receiver coil and a second receiver coil formed in the substrate and inductively coupled by electromagnetic induction caused by energization to the transmitter coil, wherein the substrate is a multilayer substrate with six or more wiring layers and insulating layers alternately arranged between each of the six or more wiring layers, and the substrate includes a plurality of conductive penetrating portions each of which is formed to penetrate at least one of the insulating layers so that the plurality of conductive penetrating portions connect the six or more wiring layers, each of the first receiver coil and the second receiver coil is wound a plurality of turns in each wiring layer, excluding at least one wiring layer, among the six or more wiring layers, and is connected through the plurality of conductive penetrating portions, and at least one of the plurality of conductive penetrating portions is disposed inside each of the first receiver coil and the second receiver coil.
2. The position detection device according to claim 1, wherein the substrate includes ten wiring layers, and each of the first receiver coil and the second receiver coil is wound a plurality of turns in each wiring layer, excluding at least one wiring layer, among the ten wiring layers.
3. The position detection device according to claim 2, wherein the plurality of conductive penetrating portions include three first conductive portions formed to penetrate five of the insulating layers, and four second conductive portions formed to penetrate one of the insulating layers, at least one of the three first conductive portions is disposed inside each of the first receiver coil and the second receiver coil, and at least one of the four second conductive portions is disposed inside each of the first receiver coil and the second receiver coil.
4. The position detection device according to claim 3, wherein the ten wiring layers are referred to as a first wiring layer, a second wiring layer, a third wiring layer, a fourth wiring layer, a fifth wiring layer, a sixth wiring layer, a seventh wiring layer, an eighth wiring layer, a ninth wiring layer, and a tenth wiring layer from one side to another side in a normal direction to a surface direction of the substrate, each of the first receiver coil and the second receiver coil is wound the plurality of turns in each of the first wiring layer, the second wiring layer, the third wiring layer, the fourth wiring layer, the seventh wiring layer, the eighth wiring layer, the ninth wiring layer, and the tenth wiring layer, excluding the fifth wiring layer and the sixth wiring layer, the three first

conductive portions formed in the substrate include: an inside third-layer-to-fourth-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the third wiring layer and the fourth wiring layer, and is disposed inside each of the first receiver coil and the second receiver coil; an outside fourth-layer-to-seventh-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the fourth wiring layer and the seventh wiring layer, and is disposed outside each of the first receiver coil and the second receiver coil; and an inside seventh-layer-to-eighth-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the seventh wiring layer and the eighth wiring layer, and is disposed inside each of the first receiver coil and the second receiver coil, and the four second conductive portions formed in the substrate include: an inside first-layer-to-second-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the first wiring layer and the second wiring layer, and is disposed inside each of the first receiver coil and the second receiver coil; an outside second-layer-to-third-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the second wiring layer and the third wiring layer, and is disposed outside each of the first receiver coil and the second receiver coil; an outside eighth-layer-to-ninth-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the eighth wiring layer and the ninth wiring layer, and is disposed outside each of the first receiver coil and the second receiver coil; and an inside ninth-layer-to-tenth-layer conductive portion that connects each of the first receiver coil and the second receiver coil formed in the ninth wiring layer and the tenth wiring layer, and is disposed inside each of the first receiver coil and the second receiver coil.
5. The position detection device according to claim 4, wherein the inside first-layer-to-second-layer conductive portion and the inside ninth-layer-to-tenth-layer conductive portion are disposed at positions overlapping with each other in the normal direction, and the outside second-layer-to-third-layer conductive portion and the outside eighth-layer-to-ninth-layer conductive portion are disposed at positions overlapping with each other in the normal direction.
6. The position detection device according to claim 1, wherein the substrate includes eight wiring layers, and each of the first receiver coil and the second receiver coil is wound the plurality of turns in each wiring layer, excluding at least one wiring layer, among the eight wiring layers.
7. The position detection device according to claim 6, wherein the plurality of conductive penetrating portions include one first conductive portion formed to penetrate three of the insulating layers, and four second conductive portions formed to penetrate one of the insulating layers, the one first conductive portion is disposed inside each of the first receiver coil and the second receiver coil, and at least one of the four second conductive portions is disposed inside each of the first receiver coil and the second receiver coil.
8. The position detection device according to claim 1, wherein the substrate includes six wiring layers, and each of the first receiver coil and the second receiver coil is wound the plurality of turns in each wiring layers, excluding at least one wiring layer, among the six wiring layers.
9. The position detection device according to claim 8, wherein the plurality of conductive penetrating portions include one first conductive portion formed to penetrate three of the insulating layers, and two second conductive portions formed to penetrate one of the insulating layers, the one first conductive portion is disposed inside each of the first receiver coil and the second receiver coil, and at least one of the two second conductive portions is disposed inside each of the first receiver coil and the second receiver coil.
10. The position detection device according to claim 6, wherein each of the first receiver coil and the second receiver coil is formed in each wiring layer, excluding two central wiring layers, among the six or more wiring layers.
11. The position detection device according to claim 1, wherein each of the first receiver coil and the second receiver coil includes a portion having a spiral pattern shape.
12. The position detection device according to claim 1, wherein one of the first receiver coil and

the second receiver coil has a pattern shape that draws a sine curve, and another of the first receiver coil and the second receiver coil has a pattern shape that draws a cosine curve.

13. The position detection device according to claim 1, further comprising a detection object configured to generate an induced current by electromagnetic induction in response to an influence of the transmitter coil that is energized, and to change induced electromotive forces generated in the first receiver coil and the second receiver coil by the induced current that is generated.

14. The position detection device according to claim 13, further comprising a signal processing unit configured to calculate a position of the detection object based on the induced electromotive forces output by the first receiver coil and the second receiver coil.